Chapter 2. Probability, Likelihood and Bayesian Estimation

Random variables, maximum likelihood, MAP estimation, Bayesian models

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1 Events and Conditional Probability

A sample space Ω contains the possible outcomes and events are (measurable) subsets. The axioms are nonnegativity, $P(\Omega) = 1$ and countable additivity for disjoint events. They imply

$$P(\emptyset) = 0, \qquad P(E^c) = 1 - P(E), \qquad P(E \cup F) = P(E) + P(F) - P(E \cap F),$$

monotonicity $E \subseteq F \Rightarrow P(E) \le P(F)$, and continuity along increasing or decreasing sequences of events.

Definition 2.1: conditional probability, Bayes' rule, independence

For $P(E) > 0$, $P(F \mid E) = P(F \cap E)/P(E)$. For a partition $E_1, E_2, \dots$ of Ω,

$$P(F) = \sum_i P(F \mid E_i)P(E_i), \qquad P(E_i \mid F) = \frac{P(F \mid E_i)P(E_i)}{\sum_j P(F \mid E_j)P(E_j)}.$$

E, F are independent if $P(E \cap F) = P(E)P(F)$.

Example 2.2: why the prior matters

Let $P(D) = 0.01$, $P(+ \mid D) = 0.9$, $P(+ \mid D^c) = 0.1$. Then

$$P(D \mid +) = \frac{0.9(0.01)}{0.9(0.01) + 0.1(0.99)} \approx 0.0833.$$

A sensitive test does not make the posterior large when the prior is small. Two fair coin tosses: "first head" and "second head" each have probability $1/2$ and their intersection $1/4$, so they are independent.

2 Random Variables and Distributions

A random variable is a measurable function $X : \Omega \to \mathbb{R}$ (e.g. the number of heads in three tosses, $X(HHT) = 2$). Its CDF is $F_X(t) = P(X \le t)$.

- Discrete: probability mass $p(x) = P(X = x)$, $F_X(t) = \sum_{x \le t} p(x)$.
- Continuous: density $f \ge 0$ with $\int f = 1$, $P(a < X \le b) = \int_a^b f(u)\,du$. A density value is not a probability and may exceed one; $P(X = x) = 0$.

$$\mathbb{E}[g(X)] = \sum_x g(x)p(x) \text{ or } \int g(x)f(x)\,dx, \qquad \operatorname{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}X)^2.$$

Theorem 2.3: tools used throughout the course

(i) Linearity: $\mathbb{E}[aX + bY] = a\mathbb{E}X + b\mathbb{E}Y$ always.

(ii) Independence: $\mathbb{E}[XY] = \mathbb{E}X\,\mathbb{E}Y$; hence $\operatorname{Var}(X+Y) = \operatorname{Var}X + \operatorname{Var}Y$. Uncorrelatedness ($\operatorname{Cov} = 0$) suffices for the variance identity.

(iii) Tower property: $\mathbb{E}[\mathbb{E}[Z \mid X]] = \mathbb{E}Z$.

(iv) $\mathbb{E}[(Z - c)^2] = \operatorname{Var}Z + (\mathbb{E}Z - c)^2 \ge \operatorname{Var}Z$ for every constant c.

Proof. (iv): write $Z - c = (Z - \mathbb{E}Z) + (\mathbb{E}Z - c)$, square, and use $\mathbb{E}[Z - \mathbb{E}Z] = 0$ to kill the cross term. The other items are standard. ∎

3 Maximum Likelihood Estimation

Definition 2.4: likelihood and MLE

For iid data $x_1, \dots, x_n$ from $p(x \mid \theta)$,

$$L(\theta) = \prod_{i=1}^{n} p(x_i \mid \theta), \qquad \hat{\theta}_{\mathrm{MLE}} = \arg\max_{\theta} \sum_{i=1}^{n} \log p(x_i \mid \theta).$$

The data are fixed and the parameter varies; L is not a distribution over θ. The logarithm preserves the maximiser and turns products into sums.

Example 2.5: Bernoulli, including the boundary

$x_i \in \{0, 1\}$ with $P(x_i = 1) = \theta$; let $s = \sum x_i$, $f = n - s$. Then $\log L = s \log \theta + f \log(1 - \theta)$. For $0 < s < n$,

$$\frac{d}{d\theta} \log L = \frac{s}{\theta} - \frac{f}{1 - \theta} = 0 \iff \theta = \frac{s}{n}, \qquad \frac{d^2}{d\theta^2} \log L < 0.$$

If $s = 0$, $L = (1 - \theta)^n$ is decreasing and the maximiser is 0; if $s = n$ it is 1. Both agree with s/n, but the derivative argument cannot be used there. Eight heads in ten tosses gives 0.8; two positive reviews out of two gives the extreme estimate 1.

Example 2.6: Gaussian

For $x_i \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$, $\log L = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum (x_i - \mu)^2$, giving $\hat{\mu} = \bar{x}$ and $\hat{\sigma}^2 = \frac{1}{n} \sum (x_i - \bar{x})^2$ (biased; the unbiased version divides by $n - 1$).

4 Bayesian Inference and MAP

Definition 2.7: prior, posterior, evidence, MAP

With prior $p(\theta)$ and likelihood $p(D \mid \theta)$,

$$p(\theta \mid D) = \frac{p(D \mid \theta) p(\theta)}{p(D)}, \qquad p(D) = \int p(D \mid \theta) p(\theta) \, d\theta,$$

$$\hat{\theta}_{\mathrm{MAP}} = \arg\max_{\theta} \{\log p(D \mid \theta) + \log p(\theta)\}.$$

The evidence does not depend on θ. MAP is the posterior mode; the posterior mean is a different summary.

4.1 Beta–Bernoulli conjugacy

Theorem 2.8: Beta prior

If $\theta \sim \mathrm{Beta}(a, b)$, i.e. $p(\theta) = \theta^{a-1}(1 - \theta)^{b-1} / B(a, b)$ with $B(a, b) = \Gamma(a)\Gamma(b)/\Gamma(a + b)$, and s

successes, f failures are observed, then

$$\theta \mid D \sim \mathrm{Beta}(a+s,\, b+f), \qquad \mathbb{E}[\theta \mid D] = \frac{a+s}{a+b+n}, \qquad \hat{\theta}_{\mathrm{MAP}} = \frac{a+s-1}{a+b+n-2}$$

(the mode formula needs both posterior parameters above one).

Proof. $p(\theta \mid D) \propto \theta^s(1-\theta)^f \cdot \theta^{a-1}(1-\theta)^{b-1} = \theta^{(a+s)-1}(1-\theta)^{(b+f)-1}$, which is the $\mathrm{Beta}(a+s, b+f)$ kernel. For $\mathrm{Beta}(\alpha,\beta)$, $\mathbb{E}\theta = B(\alpha+1,\beta)/B(\alpha,\beta) = \alpha/(\alpha+\beta)$ using $\Gamma(z+1) = z\Gamma(z)$. The mode solves $\frac{\alpha-1}{\theta} = \frac{\beta-1}{1-\theta}$, i.e. $\theta = \frac{\alpha-1}{\alpha+\beta-2}$. ■

The uniform prior is $\mathrm{Beta}(1,1)$; then the posterior mean $(s+1)/(n+2)$ is Laplace's rule of succession, and it equals the posterior predictive probability of a success, since $P(X_{\mathrm{new}} = 1 \mid D) = \int \theta\, p(\theta \mid D)\, d\theta$.

Example 2.9: the two sellers

Seller 1 has 90 positive and 10 negative reviews; seller 2 has 2 positive and 0 negative. With uniform priors

$$\theta_1 \mid D_1 \sim \mathrm{Beta}(91, 11), \qquad \theta_2 \mid D_2 \sim \mathrm{Beta}(3, 1).$$

Posterior predictive probabilities of a positive next review: $91/102 \approx 0.892$ versus $3/4 = 0.75$, so seller 1 is preferred, whereas the MLE (0.9 versus 1) prefers seller 2. A different question is which unknown rate is larger: since $\mathrm{Beta}(3,1)$ has CDF t^3,

$$P(\theta_1 > \theta_2 \mid D) = \mathbb{E}[\theta_1^3] = \frac{91 \cdot 92 \cdot 93}{102 \cdot 103 \cdot 104} \approx 0.713.$$

5 MAP as Regularisation and the Bias–Variance Decomposition

5.1 Gaussian mean with a Gaussian prior (Problem Set 1, Problem 2)

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\theta, 1)$ and prior $\theta \sim N(0, \tau^2)$.

Theorem 2.10: MAP is ridge, and it shrinks

$$\hat{\theta}_{\mathrm{MAP}} = \arg\min_{\theta} \sum_{i=1}^{n} (X_i - \theta)^2 + \lambda\theta^2, \quad \lambda = \frac{1}{\tau^2}, \qquad \hat{\theta}_{\mathrm{MAP}} = a\bar{X}, \quad a = \frac{n}{n+\lambda} = \frac{n\tau^2}{n\tau^2+1}.$$

Proof. $-\log p(\theta \mid X) = \frac{1}{2}\sum(X_i - \theta)^2 + \frac{\theta^2}{2\tau^2} + \text{const}$; multiplying by 2 gives the ridge objective. Setting the derivative $-2\sum(X_i - \theta) + 2\lambda\theta$ to zero gives $(n+\lambda)\theta = n\bar{X}$. ■

5.2 Bias–variance decomposition

Theorem 2.11: MSE = bias^2 + variance

For any estimator $\hat{\theta}$ of θ, $\mathbb{E}[(\hat{\theta} - \theta)^2] = (\mathbb{E}\hat{\theta} - \theta)^2 + \mathrm{Var}(\hat{\theta})$.

Proof. Write $\hat{\theta} - \theta = (\hat{\theta} - \mathbb{E}\hat{\theta}) + (\mathbb{E}\hat{\theta} - \theta)$. The second bracket is a constant and the first has mean zero, so the cross term vanishes after taking expectations. ■

For $\hat{\theta} = a\bar{X}$: bias $(a-1)\theta$, variance a^2/n, so $\mathrm{MSE}(a) = (a-1)^2\theta^2 + a^2/n$. Minimising over a gives $a^\star = n\theta^2/(n\theta^2+1)$, which equals the MAP shrinkage factor exactly when $\tau^2 = \theta^2$. Since $a^\star < 1$, some bias always pays off in MSE relative to the MLE $a = 1$; this is the first instance of the bias–variance trade-off that reappears for ridge regression (Chapter 4) and Adam's bias correction (Chapter 8).

Remark 2.12: Laplace prior gives the lasso

With $p(\theta) \propto e^{-|\theta|/b}$ instead, $-\log$ posterior is $\frac{1}{2}\sum(X_i-\theta)^2 + \frac{1}{b}|\theta|$ and the MAP is the soft threshold $\mathrm{sign}(\bar{X})\max(|\bar{X}| - \frac{1}{nb}, 0)$ (Chapter 4).